

Implementing Verdier's robust extrapolation in the CKT model

Relaxing the i.i.d. assumption on ν_{it}^l

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Our current estimator is Verdier's “simple” extrapolation

We currently follow Suri (2011) and Tjernström (2023):

- impose the LCA restriction $\Delta_i = \beta + \phi \theta_i$
- identify ϕ from switcher trajectories
- extrapolate to non-migrants (d_N, d_T) along the LCA line

Verdier calls our approach the *simple extrapolation* & proposes a *robust* alternative

The simple extrapolation assumes i.i.d. non-pecuniary shocks

Our current Assumption A2 states that the non-pecuniary utility shock

ν_{it}^l is i.i.d. across individuals i , time t , & locations l

This implies:

- ν_{it}^l independent of baseline skill & comparative advantage **everywhere**
- rules out systematic differences in ν_{it}^l across regions/institutions
- workers in different locations have the same shock distribution, conditional on comparative advantage

The migration literature documents what A2 rules out

Regional heterogeneity is a central fact of this literature:

- **Bryan & Morten (2019):** structural model of Indonesian internal migration estimates large bilateral migration costs
- **Lagakos et al. (2020):** migration cost estimates differ substantially across developing countries
- **Pulido & Swiecki (2021):** cross-sector barriers in Indonesia explain a large share of within-country income gaps
- **China:** hukou is an institutional source of region-specific migration barriers

Our current model rules this out by assumption

Verdier's robust extrapolation weakens A2 to a conditional version

Decompose the non-pecuniary shock:

$$\nu_{it}^l = m_l(v_i) + \tilde{\nu}_{it}^l$$

where v_i is an observed, time-invariant individual-level index

New assumption (A2'):

- $\tilde{\nu}_{it}^l$ is i.i.d. *conditional on* v_i
- $m_l(v_i)$ can correlate arbitrarily with comparative advantage

We use first-wave province as v_i to index region-level shocks (all three countries)

Why this is weaker

A2 required *unconditional* independence

A2' requires only *within-province* independence

The LCA gets a province-specific intercept

Slope stays common; intercept varies

Simple:

$$\Delta_i = \beta + \phi \theta_i$$

Robust:

$$\Delta_i = \beta(v_i) + \phi \theta_i$$

- slope ϕ remains common—this is still the LCA restriction
- intercept $\beta(v_i)$ absorbs province-level $m_l(v_i)$ heterogeneity
- ϕ identifies off within-province switcher variation
- extrapolation to non-migrants happens province-by-province, then averaged

Pooling provinces mixes differentially-selected switcher samples

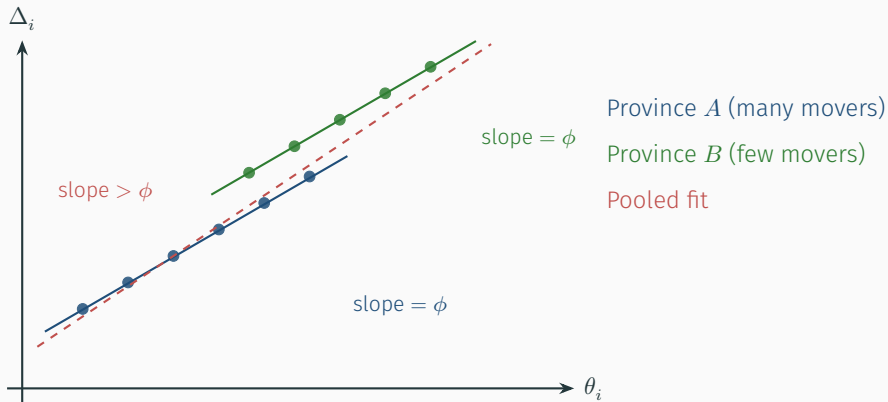
Consider two provinces:

- **Province A** (many movers): wide switcher sample, broad range of θ_i
- **Province B** (few movers): skewed sample, only high-return types migrate

Pooled regression: biased by across-province difference in selection

Within-province: clean slope, holds region-level shocks constant

Within-province slopes agree—the pooled slope is pulled off



Each province's switcher cloud has true slope ϕ

The pooled line tilts because of intercept differences, not because ϕ varies

How Suri, Verdier, and CKT
identify the LCA slope

Suri (2011) identifies ϕ via panel SUR & minimum distance

Two-stage indirect approach on a multi-year, binary-adoption panel:

Stage 1 (reduced form). Stack one outcome equation per year. SUR over the panel:

$$y_{it} = \pi_{0,t} + \sum_{\underline{d} \in \mathcal{D}} \pi_{\underline{d},t} \mathbb{1}\{\underline{d}_i = \underline{d}\} + x'_{it} \gamma_t + u_{it}$$

Each coefficient $\pi_{\underline{d},t}$ is the average y in adoption pattern $\underline{d}$ in year t

Stage 2 (structural / OMD). Express each $\pi_{\underline{d},t}$ as a known function of $(\beta, \phi, \lambda_1, \dots, \lambda_K)$ implied by the CRC restriction

$$g_{\underline{d},t}(\beta, \phi, \lambda) = \pi_{\underline{d},t}$$

Solve via weighted minimum distance: $\hat{\theta} = \arg \min (\hat{\pi} - g(\theta))' W (\hat{\pi} - g(\theta))$

- Identification: SUR over-identifies the structural parameters; ϕ pinned by the across-equation cross-restrictions
- Implemented in our `randcoef.ado`, which exactly replicates Suri's published results

Verdier identifies α_1 at the worker level

Two-step procedure on Suri's (2011) Kenyan data:

Step 1: Chamberlain (1992) projection. Within each household, partial out a separate fixed effect for treated and untreated periods:

$$y_{it} = \mu_{\text{hhid}_i, D_{it}} + x'_{it}\gamma + e_{it}$$

Recover worker-level a_i (untreated mean) and $a_i + b_i$ (treated mean)

Step 2: LCA regression at worker level.

$$a_i = \alpha_0 + \alpha_1 \cdot b_i + \varepsilon_i$$

α_1 is his ϕ

Robust identification

Instrument the endogenous b_i with *village-demeaned* period-treatment indicators
Within-village treatment variation only $\Rightarrow$ purges $m_l(v_i)$ confounders

CKT identifies ϕ at the trajectory level

Trajectory pooling: every worker with $\underline{d}_i = \underline{d}$ has the same expected baseline

$$E[\theta_i \mid \underline{d}_i = \underline{d}] = \mu_{\underline{d}} - \mu_{\underline{d}_0}$$

Single GMM equation, one scalar $\mu_{\underline{d}}$ per trajectory:

$$y_{it} = \sum_{\underline{d}} \mu_{\underline{d}} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \Delta_{\underline{d}_0} D_{it} + \phi \sum_{\underline{d} \in \mathcal{D}_S \setminus \underline{d}_0} (\mu_{\underline{d}} - \mu_{\underline{d}_0}) D_{it} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \dots$$

Identification: switcher $\times$ choice indicators as instruments

Robust variant: same equation, switcher $\times$ choice instruments *village-demeaned* (just like VV)

Asymptotically the two estimators target the same ϕ

VV's worker-level LCA:

$$a_i = \alpha_0 + \alpha_1 b_i + \varepsilon_i$$

Take expectations conditional on trajectory:

$$E[a_i \mid \underline{d}_i = s] = \alpha_0 + \alpha_1 E[b_i \mid \underline{d}_i = s]$$

Under trajectory pooling (A3), worker types are indistinguishable across clusters within trajectory:

$$E[a_i \mid \underline{d}_i = s] = \mu_{\underline{d}}, \quad E[b_i \mid \underline{d}_i = s] = \Delta_{\underline{d}}$$

so

$$\mu_{\underline{d}} = \alpha_0 + \alpha_1 \Delta_{\underline{d}}$$

which is exactly CKT's trajectory-level LCA $\Delta_{\underline{d}} = \beta + \phi (\mu_{\underline{d}} - \mu_{\underline{d}_0})$ with $\alpha_1 = \phi$

Same population ϕ . Different aggregation level

Both estimators are consistent under three assumptions

A1 LCA at the worker level: $\Delta_i = \beta(v_i) + \phi \theta_i + \xi_i$

A2' Within-cluster exogeneity: $E[u_{it}, \xi_i \mid v_i, D_{i1}, \dots] = 0$

A3 Trajectory pooling: the distribution of θ_i given $\underline{d}_i$ is the same in every cluster

Then $\hat{\alpha}_1^{\text{VV}} \xrightarrow{P} \phi$ and $\hat{\phi}^{\text{robust_vv}} \xrightarrow{P} \phi$

- Same population parameter, different sample aggregations
- Asymptotic variances differ; both achieve $\sqrt{V}$ -rate under cluster asymptotics
- When A3 fails, VV stays consistent; CKT picks up an alpha-pooling bias proportional to $\text{Cov}_s(\bar{\beta}(s), \mu_{\underline{d}} - \mu_{\underline{d}_0})$

Full proof: `docs/reviews/2026-04-25_robust-vv-equivalence-proof.md`

Our first robust implementation hit a finite-sample wall

`run_grc_robust v1`: add $|V| - 1$ cluster $\times$ choice interactions as *parameters*

$$\Delta_{\underline{d}_0} D_{it} \longrightarrow \sum_v \beta(v) \mathbb{1}\{v_i = v\} D_{it}$$

Asymptotically equivalent to VV's instrument-demeaning approach

Empirical reality: $G \in \{22, 26, 29\}$ clusters, ~ 47 moments

- cluster-robust weighting matrix is rank-deficient
- GMM objective is nearly flat in ϕ along a cluster-FE ridge
- TZA: two local minima— $\phi = -1.00$ vs $\phi = +1.25$ depending on initial value
- CHN: hung > 70 minutes on the first fit

v1 retired

Our v2 implementation moves cluster structure to the instruments

`run_grc_robust_vv`: keep CKT's parameters, swap the instruments

- Regress $\mathbb{1}\{\underline{d}_i = s\} D_{it}$ on $\mathbb{1}\{v_i\}$ *among trajectory-s workers*
- Use residuals as instruments for ϕ identification

Same parameter set as the simple GRC ($\mu_{\underline{d}}, \Delta_{\underline{d}_0}, \kappa, \gamma$); no $\beta(v)$ parameters

Why this works where v1 failed

No extra cluster-FE parameters $\Rightarrow$ no rank-deficient weighting matrix

Cluster structure enters only through which moments contribute to identification

Stable across initial values; degenerate- v test matches simple onestep to 6 decimals

We still want to check trajectory pooling empirically

Trajectory pooling collapses individual workers to one $\mu_{\underline{d}}$ per trajectory

GMM moments weight clusters by

$$w(s, v) = n_{s, v} \cdot \text{Var}(D \mid s, v)$$

Different switchers $\Rightarrow$ different cluster weights $\Rightarrow$ if $\beta(v)$ varies, the pooled ϕ inherits a tilt

Mean total-variation distance between switchers' weight profiles:

Country	mean TV	Verdier shift from simple
IDN	0.258	37%
CHN	0.038	24%
TZA	0.014	4%

Suggestive: rankings match. IDN may have residual pooling bias

The diagnostic: rebuild VV's worker-level estimator on CKT data

Port [firststage_projection.do, robust.do] verbatim:

1. Chamberlain projection in CKT: absorb (pid, D) fixed effects in OLS of y on covariates
2. Extract worker-level a_i and $a_i + b_i$ from the cell means
3. 2SLS at worker level: a_i on b_i , instrumented by village-demeaned period-treatment indicators

Compare $\hat{\phi}^{VV}$ (worker-level) to $\hat{\phi}^{robust_vv}$ (trajectory-pooled)

- **Agreement** $\Rightarrow$ trajectory pooling adds no bias; `run_grc_robust_vv` stands as primary
- **Divergence** $\Rightarrow$ trajectory pooling matters; report Verdier with caveat & VV as sensitivity

Diagnostic only—we do not propose VV's worker-level approach as the paper's main estimator

Diagnostic results: TZA agrees, IDN diverges, CHN uninformative

Country	$\hat{\phi}^{\text{simple}}$	$\hat{\phi}^{\text{robust_vv}}$	$\hat{\phi}^{\text{VV-worker}}$
IDN	−0.526 (0.10)	−0.334 (0.15)	− 1.691 (0.57)
CHN	−0.205 (0.13)	−0.155 (0.23)	− 0.598 (0.90)
TZA	−0.719 (0.12)	−0.690 (0.09)	− 0.634 (0.36)

- **TZA**: full agreement, as predicted by mean TV
- **CHN**: VV's SE (0.90) so wide that CIs broadly overlap—uninformative
- **IDN**: VV says $\phi \approx -1.7$, far more negative than -0.33 . CIs barely overlap

Pro-poor sign holds under all three approaches in all three countries

What we learn—and what is still open

Confirmed

- TZA: trajectory pooling adds no detectable bias
- All approaches agree on the sign and broad magnitude of $\phi < 0$ everywhere

Open

- IDN VV-worker SE is 4× Verdier's. Genuine weak ID? Or trajectory-pooling bias?
- Multi-switch trajectories may add noise to the worker-level Chamberlain projection
- First-stage F for VV-worker not yet checked

Provisional decision

- **run_grc_robust_vv** as primary across all three countries
- Add VV-worker as a sensitivity column in the main table
- Methods note: trajectory pooling is consistent under standard assumptions; the IDN flag is unresolved

The move buys a weaker assumption at tractable cost

We gain:

- $A2'$ matches what the migration literature documents
- the CHN hukou split becomes a principled special case ($v_i = \text{hukou}$) rather than an ad hoc fix
- stayer ATE estimates defensible against Verdier's concern in Suri's data

We give up:

- Δ_{d_N} averages over provinces with switcher support, not the full population
- thinner identification of ϕ : only within-province switcher variation contributes
- cluster-robust standard errors at the v_i level

An observed shifter w_i provides a separate falsification test

Regress estimated θ_i^* on Δ_i and an observed shifter w_i :

$$\theta_i^* = \alpha_0 + \alpha_1 \Delta_i + \alpha_2 w_i + e_i, \quad H_0 : \alpha_2 = 0$$

Kenya: w_i = distance to nearest seed seller; rejects if $\alpha_2 \neq 0$

w_i must be observed, exogenous to skill conditional on Δ_i , and correlated with ν_{it}^l

Candidates in our data:

- distance / travel time to nearest urban center
- rainfall variability or pre-period climate shock (mergeable via GPS/admin code)
- distance to family already in urban area (closest analogue, but reflection problem)

None is perfect; data availability is the binding constraint